9 November 2026

Dear Editors,

Please consider our manuscript, “Tidal potential measured with distant optical clocks”, for publication as an Article in *Nature*.

Optical clocks measure gravitational potential differences through their relative ticking rates. Using them to study changes in Earth’s gravity field requires distinguishing overlapping signals and understanding how clock noise and interrupted observations limit the precision of each component. Astronomical tides provide a predictable signal for addressing both questions.

We compare a strontium clock in coastal Shanghai with an ytterbium clock in inland Wuhan across a 692-km baseline, connected by a 1,350-km fibre link. Five campaigns provide 545.9 hours of observations spanning 353 days. The measured response to the combined solid-Earth and ocean-loading prediction is 0.96 ± 0.11, and the lunar semidiurnal group response is 1.00 ± 0.13, with statistical uncertainties of one standard deviation. Conditional spectral analyses distinguish contributions near the principal lunar and solar semidiurnal frequencies. Robustness tests assess the lunar response’s sensitivity to specified daily disturbances and ocean-loading assumptions; the solar-frequency contribution remains potentially confounded with laboratory variations at the same frequency.

We develop a signal-to-noise model that combines signal strength, clock noise and actual sampling. It closely estimates the measured precision of the lunar component and quantifies how observing duration, phase coverage and parameter coupling limit the weaker components. We discuss residual contamination through sampling aliases and non-white laboratory variations as possible reasons for the poorer agreement for these components.

The study connects quantum metrology with gravitation and geophysics by showing how distant clocks can identify components of a time-varying gravitational signal and assess their attainable precision. Characterizing this tidal background is relevant to future optical-clock networks and their use in timekeeping and geodesy.

The 274-hour summer record is shared with the absolute frequency-ratio study by Hou and colleagues (arXiv:2610.34250), as disclosed in the manuscript. The present work uses the extended record for tidal-component analysis. An independent study has also reported tidal potential variations in a European optical-clock network: K. Stahl et al., *Optical clock network tracks tidal variations of the gravity potential*, arXiv:2610.39635v1 (2026).

Thank you for considering our manuscript. Our suggested referees are listed on the following page.

Yours sincerely,

[Corresponding author name, institution and email to be confirmed]

On behalf of the authors

For your reference, we would like to recommend the following experts as potential referees:

- Jun Ye (jun.ye@colorado.edu) for his pioneering work in optical clocks and frequency ratios.
- Tara M. Fortier (tara.fortier@nist.gov) for her pioneering work in optical clock comparisons and the re-definition of the SI second.
- Sébastien Bize (sebastien.bize@obspm.fr) for his pioneering work in optical frequency standards and the redefinition of the SI second.
- Helen S. Margolis (helen.margolis@npl.co.uk) for her pioneering work in optical frequency standards and clock comparisons.
- Laura C. Sinclair (laura.sinclair@nist.gov), National Institute of Standards and Technology (NIST), Boulder, CO, USA, for her work on free-space optical links and optical clock comparisons.
- M. D. Barrett (phybmd@nus.edu.sg), National University of Singapore, Singapore, for his work on optical frequency standards, systematic uncertainty evaluation and lutetium-ion clock comparisons.